

Dr. Fabian Grünewald studied Chemistry at the University of Groningen following the Erasmus Mundus Master's program in Theoretical Chemistry and Computer Modeling. Supported by the Erasmus Mundus Scholarship awarded by the European Union, he completed research stays in France, Spain, and Belgium, after which he graduated from the program in 2018 with *summa cum laude*. He then joined the Molecular Dynamics group in Groningen for his PhD, working on the development of coarse-grained (CG) simulation methods for (bio)materials. His doctoral thesis, *Material Design using Martini*, received the highest distinction awarded to a PhD thesis in the Netherlands. In 2023, he was awarded the Independent Postdoctoral Fellowship of the Heidelberg Institute for Theoretical Studies to combine high-throughput CG simulations with deep learning techniques to address design challenges in next-generation nanomedicine technologies. In the same year, he was appointed core developer of the Martini Force Field Initiative. As part of the SCALE Cluster of Excellence - SubCellular Architecture of Life, he became an independent group leader at FIAS in 2026, leading the *Virtual Integrative Platforms for Synthetic Biology* group. Using digital twins built from simulations and machine learning models, his group investigates the molecular architecture and underlying functional principles of biomimetic systems.

Links:

Google Scholar: <https://scholar.google.com/citations?user=rvU2n2AAAAAJ&hl=en>

ORCID: <https://orcid.org/0000-0001-6979-1363>